

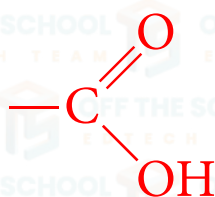
UNIT 09

CARBONYL COMPOUNDS I: CARBOXYLIC ACID & FUNCTIONAL DERIVATIVES

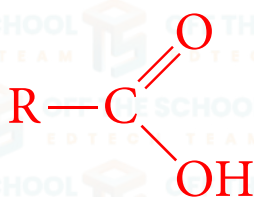


CARBOXYLIC ACID:

"Organic compounds which contains Carbonyl group (C=O) attached to hydroxyl group (–OH) are referred to as carboxylic acid."



carboxyl group



or RCOOH or RCO_2H

three ways to represent a carboxylic acid

1. What is a Dimer?

• A dimer is a molecule formed when two molecules of the same substance join together. For carboxylic acids, these dimers are often formed through a specific type of bonding called hydrogen bonding.

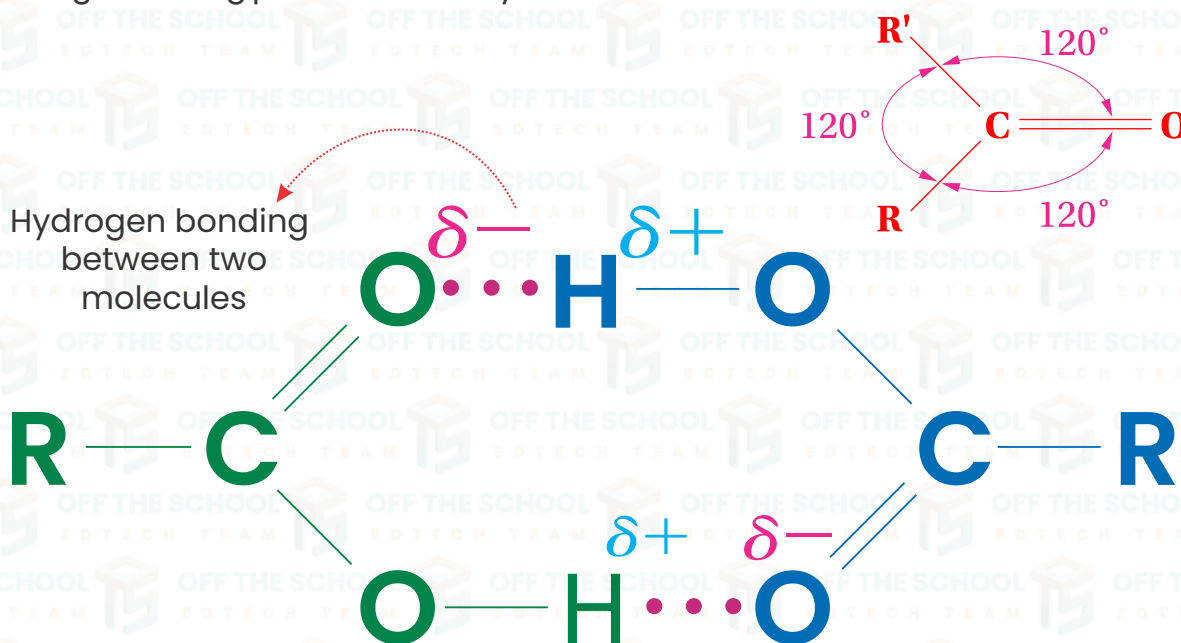
2. Hydrogen Bonding in Cyclic Dimers:

• In cyclic dimers of carboxylic acids, the hydrogen atom from the –OH (hydroxyl) group of one acid molecule forms a bond with the oxygen atom of the –COOH (carboxyl) group of another acid molecule. This creates a ring-like structure, hence the term "cyclic" dimer.

3. Importance of Cyclic Dimers:

• The formation of cyclic dimers is significant because it influences the physical properties of carboxylic acids, such as their boiling points and solubility. Carboxylic acids have higher boiling points compared to other organic compounds of similar molecular weight, mainly due to the stability provided by these dimeric structures.

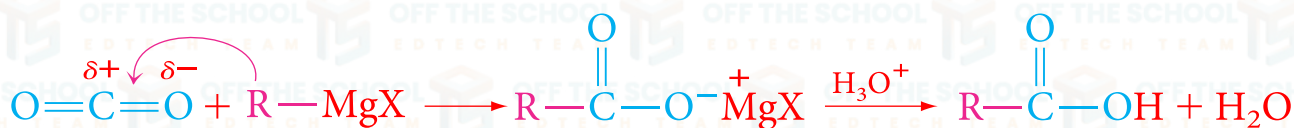
• Acetic acid, the acid that gives vinegar its sour taste, forms cyclic dimers. The formation of these dimers contributes to the distinct properties of vinegar, including its boiling point and acidity.



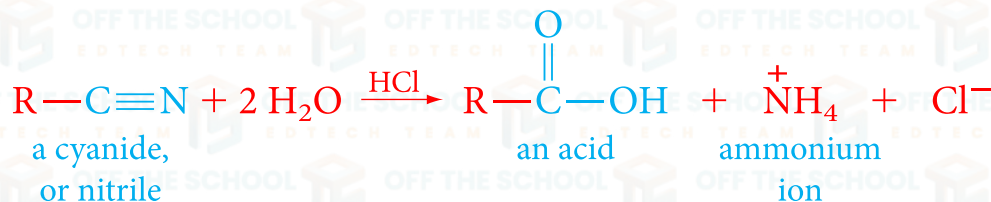
hydrogen-bonded acid dimer

Preparation:

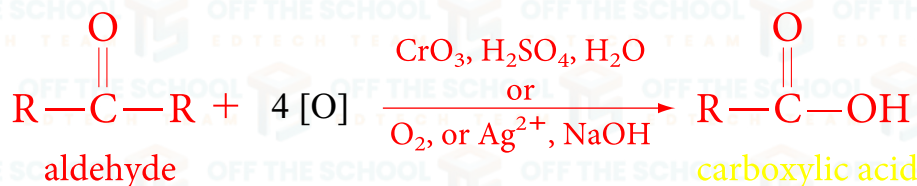
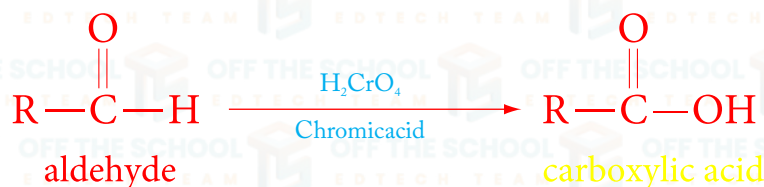
From Carbon dioxide or Carbonation of Grignard's reagent



By Hydrolysis of Nitriles

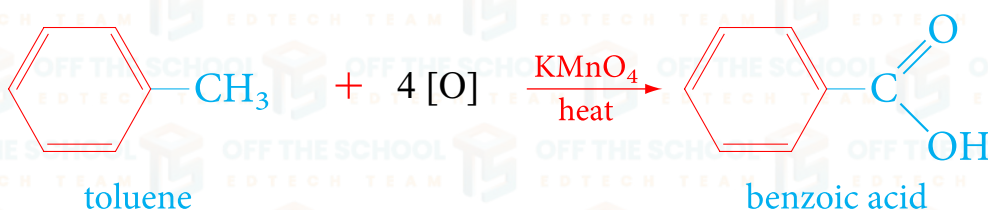


Oxidation of Aldehydes & Ketones



Oxidation of Aromatic side chains

Aromatic acids can be prepared by oxidizing an alkyl side chain on an aromatic ring.

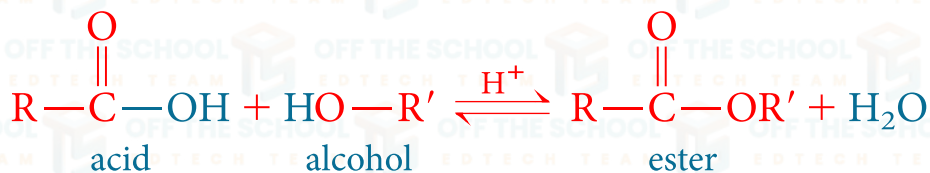


CONVERSION OF CARBOXYLIC ACIDS

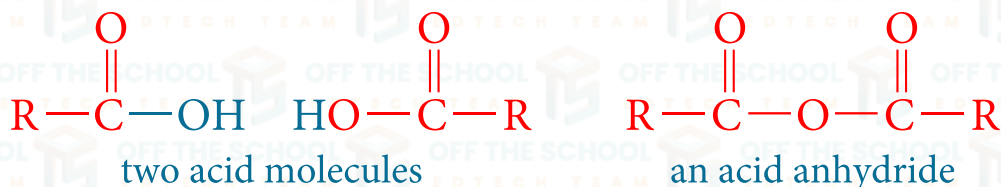
Into acyl halide



Into Esters



Into acid anhydrides



Aliphatic Carboxylic Acids

Carbon atoms	Formula	Source	Common name	IUPAC name
1	HCOOH	ants (Latin, <i>formica</i>)	formic acid	methanoic acid
2	CH ₃ COOH	vinegar (Latin, <i>acetum</i>)	acetic acid	ethanoic acid
3	CH ₃ CH ₂ COOH	milk (Greek, <i>protos pion</i> , first fat)	propionic acid	propanoic acid
4	CH ₃ (CH ₂) ₂ COOH	butter (Latin, <i>butyrum</i>)	butyric acid	butanoic acid
5	CH ₃ (CH ₂) ₃ COOH	valerian root (Latin, <i>valere</i> , to be strong)	valeric acid	pentanoic acid
6	CH ₃ (CH ₂) ₄ COOH	goats (Latin, <i>caper</i>)	caproic acid	hexanoic acid
7	CH ₃ (CH ₂) ₅ COOH	vine blossom (Greek, <i>oenanthe</i>)	enanthic acid	heptanoic acid
8	CH ₃ (CH ₂) ₆ COOH	goats (Latin, <i>caper</i>)	caprylic acid	octanoic acid
9	CH ₃ (CH ₂) ₇ COOH	pelargonium (an herb with stork-shaped seed capsules; Greek, <i>pelargos</i> , stork)	pelargonic acid	nonanoic acid
10	CH ₃ (CH ₂) ₈ COOH	goats (Latin, <i>caper</i>)	capric acid	decanoic acid